

Seoyeon Park

Aspiring Software Engineer

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CAREER OBJECTIVE

A motivated and passionate aspiring software engineer pursuit of a Bachelor's Degree in Computer Engineering, aiming to leverage academic knowledge and hands-on project experiences to secure an internship. Eager to apply theoretical knowledge in a practical environment to develop real-world software solutions, enhance technical skills. Looking forward to gaining hands-on experience as **Backend Developer** and **ML Engineer**.

EDUCATION

University of California, Berkeley	August 23 – May 24
Hongik University	March 21 – February 26
Computer Engineering major	3.86 / 4.5 GPA
Coursework: Computer Security, Database System, Artificial Intelligence, Operating System, Data Structures and Programming, Computer Architecture, Assembly Language Programming, Algorithms and Analysis, Computer Network, Object-Oriented Programming, C-Programming, Discrete Math and Probability	
NIPA-Google ML Bootcamp	October 24 – November 24
Google for Developers Machine Learning Bootcamp	July 24 – October 24
NAVER Boostcamp Challenge (Web Fullstack)	July 24 – August 24

PROJECT EXPERIENCE

Real-Time Multimodal AI System for Smart Emergency Management	October 24 – December 24
- Developed and implemented a real-time emergency detection system from scratch, integrating audio anomaly detection with audio model and contextual interpretation with multimodal model. - Built a robust detection system with a FastAPI-based server, a Kotlin-powered demo app, and integrated models, including the audio model with CNN (utilizing MFCC and Librosa) and the multimodal model with Transformers (utilizing OpenCV, MediaPipe, Keras, TensorFlow, and Numpy).	
BYOW: Build Your Own World	January 24 – May 24
- Designed and developed a pseudo-random world generation and interactive game with a terminal launchable GUI in Java, integrating a Tile Engine for procedural map creation and StdDraw library for real-time user interaction. - Implemented advanced features such as customizable avatars and win/lose mechanics to enhance user engagement, allowing players to select avatars and play with clear objectives.	
RookieDB: Advanced, Concurrent Database	January 24 – May 24
Designed and crafted database systems in Java, integrating advanced features such as B+ tree indices, efficient join algorithms, and query optimization, while enabling concurrent transaction execution through multigranularity locking and ensuring robust data integrity with comprehensive recovery mechanisms.	
PintOS: Efficient, Secure Operating System	August 23 – December 23
- Configured Docker-based virtual environment to streamline testing, debugging, and deployment of core functionalities. - Designed and implemented core functionalities from scratch in C and Assembly, implementing user processes, protected address spaces, and multi-threading based on priority scheduler, synchronizing operations with semaphores and locks. (unit testing with Perl)	
Pacman/Ghostbusters: AI-Powered Game	August 23 – December 23
Developed and optimized core algorithms for Pacman and Ghostbusters games in Python, utilizing search algorithms, CSPs, and MDP/RL for strategic decision-making. Applied Bayesian Networks and HMMs to model uncertainties and improve behavior prediction, enhancing overall game performance.	

CERTIFICATES

Deep Learning Specialization	issued by Coursera July 24
Microsoft Certified: Azure AI fundamentals	issued by Microsoft January 23

TECHNICAL SKILL

Programming Languages: Python, Java, Spring, Golang, C, C++, HTML, CSS, JavaScript, SQLite, noSQL, x86, MIPS, Linux
Developer Tools and Frameworks: Spring Boot, Spring Framework, Fast API, Keras, Tensorflow, OpenCV, NumPy
IDE and Other Tools: VS Code, IntelliJ IDEA, Anaconda, GitHub, Jupyter Notebook, Docker, Slack, Discord
Languages: English(bilingual proficiency), Korean(native)